3.13. a) $T_{tq} = \begin{bmatrix} R_{tq} & p_{tq} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & \cos 90^\circ & -\sin 90^\circ & 6 \\ 0 & \operatorname{sen} 90^\circ & \cos 90^\circ & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$T_{tq} = \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 0 & -1 & 6 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{tc} = \begin{bmatrix} \cos(-90^\circ) & -\sin(-90^\circ) & 0 & 4 \\ \operatorname{sen}(-90^\circ) & \cos(-90^\circ) & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 4 \\ -1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{cm} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 45 & -\sin 45 & 2 \\ 0 & \sin 45 & \cos 45 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 2 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{tm} = T_{tc} \cdot T_{cm} = \begin{bmatrix} 0 & 1 & 0 & 4 \\ -1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 2 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{tm} = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 6 \\ -1 & 0 & 0 & 3 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b) $p_{tq} = (5,6,3) \rightarrow p_{cq} = ?$

$$p_{cq} = T_{ct} \cdot p_{tq} = T_{tc}^{-1} \cdot p_{tq}$$

$$= \begin{bmatrix} 0 & -1 & 0 & 3 \\ 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 3 \\ 1 \end{bmatrix}$$

$p_{cq} = (-3,1,3)$ ✓

c) $p_{tc} = (5,6,0) \rightarrow p_{cc'} = T_{tc}^{-1} \cdot p_{tc}$

$$p_{cc'} = \begin{bmatrix} 0 & -1 & 0 & 3 \\ 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix} = (-3,1,0)$$ ✓

$$T_{ct} = T_{tc}^{-1} = \begin{bmatrix} R_{tc}^{-1} & P_{ct} \\ 0 & 1 \end{bmatrix} \rightarrow T_{cc'} = \begin{bmatrix} R_{tc}^{-1} & P_{cc'} \\ 0 & 1 \end{bmatrix}$$

$$T_{cc'} = \begin{bmatrix} 0 & -1 & 0 & -3 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

✓